

MICROPULSE

Analytics Views — Coach Guide

How to read every monitoring view, what each one is for, and when to use it

1. Overview — How the views fit together

MicroPulse splits player monitoring into a few coherent views. Each one answers a different question and uses different signals. The dashboard is built so that a coach starting their morning at the Today tab sees the verdict for every player, then drills into specific intelligence views when they need to understand WHY a verdict came out the way it did, or when they want to spot risks the daily verdict might miss.

Where each view sits in the workflow:

View	Question it answers	Frequency
Today Command Center	Who can train fully today, who needs modification, who is out?	Every morning, every staff
Decision Summary modal	WHY does this player have THIS verdict? What should I tell them?	Click into a flagged player
Indoor Load Intelligence	Indoor-mode load tracking. How heavy was the indoor session vs this player's normal?	Winter season, hall sessions
Decel Intelligence (McBurnie)	Is this player at risk from accumulated braking load? Are they coupling decels with their sprints?	Weekly review, contact-sport seasons
Quadrant View (Gabbett)	Which players are in the high-injury-risk quadrant (high external + high internal load)?	Weekly planning, trend tracking
Injury Pattern Analysis	When players got hurt, did MicroPulse flag warning signs beforehand? Proof-of-ROI.	Monthly retrospective, sales evidence
RTP Tab (in dashboard)	Where is this injured player in their return-to-play protocol? Cleared yet?	Daily during rehab

All views read from the same Catapult / RPE / VALD / WHOOP signals — they just slice the same data with different questions in mind.

2. How a Verdict Is Computed (Integration Model)

Every per-player verdict on the dashboard goes through a single decision builder — `buildAthleteDecision()` in `src/lib/micropulse/domain/decision/buildDecision.ts`. The function takes signals from every intelligence layer and combines them into one outcome (`athleteState`: GREEN / YELLOW / RED / GRAY, plus a `sessionMode` and `loadAction`).

This section explains exactly which views feed the verdict and which are standalone analytics. Important to read before the per-view sections below — knowing the integration model tells you how seriously to treat what you see in any single view.

2.1 The severity ladder

Each input signal maps to one of four severity levels. The function then applies gates: any RED gate returns RED; any YELLOW gate (when no RED gates fire) returns YELLOW; otherwise the readiness state passes through. This means ANY intelligence layer can escalate a player — there is no "averaging away" of a real concern.

Signal source	RED gate	YELLOW gate
Hard block / Rehab / RTP	Any active block	—
Readiness Decision (STEN, z-score, delta-z, volatility)	RED state from readiness engine	YELLOW state
Neural Load (CMJ + autonomic + neural composite)	suppressed (CRITICAL/HIGH)	caution (RISING)
Injury Risk (VALD + GPS spike + soreness)	HIGH	MODERATE
Composite Load Concern (Outdoor: RPE+NBS+Metabolic)	high	moderate
Indoor Composite Score (FMP / IMU höll-mode)	spike (≥ 140)	heavy (110–140)
Indoor McBurnie ratio	red (< 0.5 or > 15)	yellow (0.5–1 or 10–15)
Indoor ACWR (Gabbett 2017)	red (< 0.5 or > 1.5)	yellow (0.5–0.8 or 1.3–1.5)
Decel Intelligence (McBurnie 2022 4-dim overall)	red	yellow

If any RED gate fires, athleteState = RED. Otherwise, if any YELLOW gate fires, athleteState = YELLOW. Otherwise, the readiness state passes through unchanged.

2.2 Views that are standalone analytics (do NOT move the verdict)

Two views deliberately stay out of the verdict pipeline because they answer different questions:

- Quadrant View — visualisation only. Plots external $\times$ internal load on a 2D chart. The signals it uses (avg distance, avg sRPE) ALREADY feed the verdict via Composite Load Concern and Readiness Decision. The chart itself is a perspective on the same data, not a separate input.
- Injury Pattern Analysis — retrospective only. Computes pattern-match scores from PAST injuries against the 14-day signal window before each. Used for proof-of-ROI and methodology refinement, not for the verdict on today's players. Today's verdict reads from the live RTP tab data via injury status (which IS in the verdict gates).

2.3 Why every monitoring view feeds the verdict

Until April 2026, the verdict only used Readiness + Injury + Composite Load (outdoor) + Neural + WHOOP. Indoor Composite, Indoor McBurnie, and Decel Intelligence were computed but only surfaced in the explanation layer (the Why text shown in the player modal) — they did NOT actually move the verdict.

This was a real gap. A player flagged red on /coach/indoor-load could still show GREEN on the dashboard. A player with all four McBurnie deceleration dimensions in the danger zone could still show GREEN if their check-in was OK. Coaches drilling into intelligence views had to mentally combine what they saw with the dashboard verdict.

After April 2026 wiring, all of these layers feed buildAthleteDecision through the new indoorLoad and decelIntelligence parameters, with the same severity ladder as the existing

loadConcernLevel. The verdict now reflects every view a coach can drill into, and the dashboard can never disagree with the intelligence pages.

2.4 Why-line transparency

When an intelligence layer escalates the verdict, its summary is appended to the verdict's Why lines (buildDecisionReasons). For example: "Indoor composite is in SPIKE band — overload risk if sustained" or "Indoor McBurnie ratio in RED — decel:intensity coupling significantly off". The coach reading the modal sees both the verdict (RED) and which intelligence layer drove it.

2.5 The full pipeline (signals → verdict)

Step-by-step what happens when the dashboard renders one player's verdict:

- 1. Fetch raw signals: check-in, Catapult GPS daily, indoor session RPC, McBurnie decel RPC, RPE entries, injury status, VALD, WHOOP
- 2. Run Readiness Decision → readinessState (GREEN/YELLOW/RED/GRAY) + score + STEN + Why lines
- 3. Run Injury Risk Decision → injuryRiskLevel (HIGH/MODERATE/LOW) + Why lines
- 4. Run computeCompositeLoadConcern (outdoor: RPE+NBS+Metabolic + safety nets) → concernLevel + summary
- 5. Fetch Indoor Load Intelligence (get_indoor_load_status RPC) → composite_band + mcburnie_flag + acwr_flag
- 6. Fetch Decel Intelligence (get_mcburnie_decel_status RPC) → overall_flag (when wired in section 11.5 below)
- 7. Classify Neural Load → status (clear/caution/suppressed)
- 8. Pass everything to buildAthleteDecision → athleteState + sessionMode + loadAction + reasons
- 9. Pass athleteState through resolveDisplayAction → final FULL/MODIFIED/RECOVERY display action
- 10. Render the modal: explanation engine (buildVerdictExplanation) generates Why / Coach guidance / Verify / Ask based on action + readiness + indoor + outdoor signals

2. Today Command Center

Landing view on /coach. The single morning briefing every staff member sees.

2.1 What it shows

- Daily Briefing card at the top — date, day-state badge (NORMAL DAY / ACTIVATION / MD-1 / etc.), one-sentence verdict ("All players ready for full program today" or "3 players need a lighter session"), top concerns list
- Team status counters — 6 tiles: FULL, REDUCED (modified), RECOVERY (protected), DOMINANT FATIGUE, NEURAL LOAD, NEXT-DAY RISK
- Volatility / Avg Risk / Protected count / Baseline players row
- Coach recommendation banner — auto-generated from the team aggregate. Either green ("Run plan as normal") or amber ("Modify session — N players flagged")
- Sync Catapult + Readiness Risk Report + Post-Training Report buttons in the header

- Decision Summary list — every player as a card, sorted with concerns first. Click a card to open the per-player modal

2.2 How to read it

Read the Daily Briefing first — the one-sentence verdict at the top is the coach's morning take-away. If "all clear", scan the team status tiles to make sure no individual is flagged. If concerns exist, click directly into Top Concerns names to open their player modal.

Tile colours follow a traffic-light convention: emerald = full availability, amber = modified, rose = recovery / protected. Numbers are players, not percentages.

2.3 Action layer

The Coach Recommendation banner is auto-generated. Coaches can override per player from the Decision Summary modal — see section 3.

Tip: If everyone is in the FULL tile and no Top Concerns exist, you can stop reading here. Spend 30 seconds and move on.

3. Decision Summary Modal (per-player verdict)

Opens when you click a player on the dashboard. The most-visited drill-down view in MicroPulse.

3.1 What it shows

- Player name and verdict pill — Ready / Lighter session / Rest / Sick / Injured / Rehab / RTP
- Coach override banner if the verdict was manually changed (red = critical override, amber = caution)
- Active injury / illness card — body part, severity, RTP stage if applicable, estimated return
- WHY block — the explanation engine output. Up to 4 reason lines.
- COACH GUIDANCE — what to physically tell the player and do in session today
- VERIFY — what to check / watch for during warm-up and first 15 min
- QUESTION FOR PLAYER — direct check-in question (e.g. "How did you sleep?")
- Primary driver line — what pushed the verdict (e.g. "well below normal, sharply declining trend")
- Baseline Readiness STEN (1–10 with band label: Very low / Below baseline / Normal / Above baseline / Peak form)

3.2 The three FULL paths

When the verdict is "Ready", the explanation engine picks one of three branches based on context:

Branch	Trigger	Coach guidance tone
Genuine FULL	All signals within healthy band, STEN ≥ 5 , no composite concern	"No restrictions — full training, sprint work OK, max-intensity efforts allowed"
Override-aware FULL	STEN ≤ 4 but coach manually set FULL	"Light start, modified-friendly. Watch warmup. Pull back if signs of fatigue."
Watchful FULL	STEN OK but composite concern	"Full session OK today, but treat as

	HIGH/MODERATE or PL spike or fatigue type flagged	borderline. Skip highest-intensity blocks. Watch volume across the rest of the week."
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All three are surfaced honestly — coach knows whether "Ready" means "truly all clear" or "OK but watch this player".

3.3 Source-aware composite line

The composite number in the Why section is labelled with its source:

- "Indoor composite N (typical, ACWR 0.90)" — FMP-derived, when team is in höll-mode
- "GPS load concern: BAND, PL N× baseline" — Catapult GPS-derived, when team is in outdoor mode

The pipeline is auto-detected per player based on recent session counts, or pinned by team-level Training Mode toggle in Settings.

4. Indoor Load Intelligence (/coach/indoor-load)

Höll-mode tracking. Built on the Football Movement Profile (FMP) so it works without GPS.

4.1 Purpose

Outdoor monitoring relies on GPS — sprint distance, max velocity, HIR distance. Indoor halls have no GPS signal, so those numbers are zero or unreliable. Indoor Load Intelligence uses Catapult's IMU-only outputs (Player Load, IMA, FMP stride bands) to give the same kind of session-by-session magnitude tracking for hall sessions.

4.2 Auto-detection

A session counts as indoor if velocity_band6 distance < 50m AND fmp_total_duration_s > 600s (10 min). This excludes outdoor sessions where players naturally hit sprint zone, and surfaces hall sessions where the IMU did meaningful work.

4.3 Layout

- Team status banner — "X ready, Y need lighter" + chip counts (FULL / lighter / rest / injured / sick)
- Top Concerns list — players needing coach attention today, sorted by severity
- Team Heatmap — 14 days × N players grid, each cell coloured by composite score band, red border on indoor sessions, faint dots on rest days. Scan in 5 seconds to spot sustained spikes vs distributed load.
- Per-player rows — expand to see Composite Score banner, 14-day sparkline, FMP movement bars, latest session stats, 28-day baseline, 7-day cumulative, McBurnie ratio, ACWR

4.4 Composite Indoor Load Score (0–150+)

Weighted average of 3 IMU-native components, normalised against the player's 28-day personal baseline. 100 = exactly an average session for that player.

Component	Weight	Source
Player Load	40%	IMU accelerometer integration
FMP Dynamic High %	33%	FMP stride bands
IMA Total	27%	IMA accel + decel events

HMLD and Decel B2-3 are not in the composite — Catapult derives those with GPS context, so they're noisy indoors. Both are still shown as session-context fields and feed the McBurnie indoor ratio.

4.5 Composite bands

Score	Band	Colour	Meaning
< 60	Light	Slate	Recovery / walkthrough
60-89	Below avg	Sky	Mild tactical
90-110	Typical	Emerald	Standard training
110-140	Heavy	Amber	Match-style training
> 140	Spike	Rose	Overload risk if sustained

4.6 ACWR (Acute:Chronic Workload Ratio)

Per Gabbett 2017. Weekly cumulative Player Load divided by 28-day weekly average. Sweet spot 0.8–1.3 (player adapted to current load). Spike above 1.5 is associated with 2–4× higher injury risk.

4.7 McBurnie Indoor Ratio

Decel B2-3 count divided by (FMP duration × Dynamic High %). Measures how many high-intensity decelerations a player performs per minute of high-intensity movement. Healthy band 1–10. < 0.5 or > 15 → at-risk: significant mismatch between brake-work and movement.

4.8 When to use it

- Daily during winter season (höll-mode dominant)
- When a player flags red on the dashboard and you need to see the magnitude story
- Weekly review — scan the Team Heatmap to see who has been sustained-heavy or distributed-spiky

5. Decel Intelligence (/coach/decel-intelligence)

McBurnie 2022 framework. Four-dimension assessment of deceleration load and risk.

5.1 Why deceleration matters

Decelerations are biomechanically distinct from accelerations: higher ground reaction forces, eccentric muscle contraction, and tissue damage potential. McBurnie, Harper, Jones & Dos'Santos (2022, Sports Medicine) showed that cumulative deceleration load correlates with

overuse injury in a 2–4 week window — making it a leading indicator that pure accel- or distance-based monitoring misses.

5.2 The four dimensions

Dimension	What it measures	Red flag
Overload exposure	Cumulative 28-day decel count vs personal daily baseline	Sustained accumulation over recent weeks
Underload risk	Recent 7-day vs match-day demand	Player isn't doing enough decels in training to handle match
Decel:Accel coupling	Eccentric vs concentric balance	Eccentric-dominant (< 0.7 ratio) → hamstring/calf strain risk
Decel:Sprint coupling	Must brake after sprinting	Sprints without brake-work → quality issue, neuromuscular fatigue
Exposure concentration	Is load distributed or concentrated on one day?	Single-day spike concentrating > 35% of 28-day total

5.3 Layout

- Team status banner with overall flag distribution (red / yellow / green / unknown)
- Per-player rows sorted by overall flag — red first, then yellow, then green
- Each row expands to show all 4 dimensions with their individual flags + explanation
- Refresh Baselines button — recomputes 28-day baselines from latest data

5.4 When to use it

- Weekly review (e.g. Mondays after weekend match)
- Pre-camp planning — spot players in Underload risk who need decel exposure before high-intensity blocks
- After running a heavy COD-focused session — check if anyone tipped into Overload exposure
- When investigating a soft-tissue injury — was decel coupling already off in the 2–4 weeks prior?

6. Quadrant View (/coach/quadrant)

Gabbett 2017 / Vanrenterghem 2017. Plots every player on a 2×2 chart of external load (X = avg distance) vs internal cost (Y = avg sRPE).

6.1 The four quadrants

Quadrant	Position	Meaning	Action
INJURY RISK	Top-right	High external + high internal — paying high cost for high work	Reduce load in next session
DECOUPLED	Top-left	Low external + high internal — paying high cost for low work	Early-warning fatigue or illness signal (Halsen 2014)
PEAK FITNESS	Bottom-right	High external + low internal — high work, low cost	Player is at peak form

UNDER-STIMULATED	Bottom-left	Low external + low internal — minimal load both sides	Not at injury risk, but maybe not getting enough stimulus
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6.2 Time window selector

Three windows: 7 days (current micro-cycle, default), 14 days, 28 days. The chart re-aggregates external load and internal cost over the chosen window. 7-day is most actionable for week planning; 28-day shows seasonal positioning.

6.3 Dot size = ACWR

Each player's dot is scaled by their ACWR (acute 7-day vs chronic 28-day). Larger dot = ratio above 1.3 (acute work outpaces chronic). A player in the Peak Fitness quadrant with a large dot is technically peaking but accumulating risk fast.

6.4 When to use it

- Weekly planning — see who is where, plan session intensity per group
- Quick triage — who is in DECOUPLED right now? That's the early-warning signal nobody else sees.
- Pre-match — a cluster of players in INJURY RISK before MD is a flag to back off Friday's session

7. Injury Pattern Analysis (/coach/injuries)

Read-only analytics view. Retrospective scoring of every recorded injury against the 14 days of MicroPulse signals that preceded it.

7.1 Purpose — proof-of-ROI

When a player gets injured, this view answers: "Did MicroPulse warn us beforehand?" The system scans the 14 days before the injury date and computes a `pattern_match_score` based on how many warning signals were active (yellow/red flags, decoupling alerts, ACWR spikes). The team-level summary shows what fraction of injuries were preceded by warnings.

7.2 Layout

- Header card — "X of Y injuries (Z%) preceded by warning signs" + breakdown by body part (Hamstring / Knee / Ankle / Groin) and average pattern score
- Logged injuries list (latest 200) — each row shows date, player, injury type, mechanism, severity
- Each row has a "✓ Preceded by warning" tag and a % match badge
- Click a row to expand: yellow days, red days, decoupling alerts, ACWR, first warning days before injury, dominant signals seen, notes

7.3 Data source — single source of truth

All injuries logged via the RTP tab automatically appear here. A Postgres trigger mirrors `player_injuries` inserts into `injury_events`, then a second trigger computes the retrospective

signal scan. No manual logging in this view — it's analytics-only. Header has an "Open RTP tab →" link to the logging surface.

7.4 When to use it

- Monthly retrospective with medical staff — review which injuries were predictable and which weren't
- Sales / pilot conversations — "Of N injuries last 365 days, MicroPulse flagged warning signs for M (P%)"
- Methodology refinement — patterns of "No prior signal" injuries point to what the verdict engine still misses

8. RTP Tab (in Dashboard, More dropdown)

Clinical injury logging + 5-stage Return-To-Play workflow.

8.1 Purpose

The single write surface for injury data. Coaches log new injuries here, track each player through the 5-stage RTP protocol, and mark cleared when ready for full availability. Injury Pattern Analysis (section 7) reads from this — they share data via Postgres trigger.

8.2 The 5 RTP stages

Stage	Phase	Criteria summary
0	Rest	POLICE protocol, pain management, no physical activity
1	Light cardio	Walking / cycling / swimming, < 70% HRmax, no pain
2	Sport-specific load	Linear running, no contact, ROM ≥ 90%, strength ≥ 70% of healthy side
3	No-contact training	Tactical work, agility, COD, full ROM, eccentric strength balance ≥ 90%
4	Full training	Full contact training with squad, isokinetic testing < 10% asymmetry, hop tests ≥ 90%
5	Match	Cleared — all stage 4 criteria sustained, return-to-match progression

8.3 What it shows

- Active / All toggle — filter to currently injured players or full history
- Per-injury card with body part, type, severity, status, current RTP stage, estimated/actual return dates
- "Advance to: [next stage]" button on each active injury for quick stage progression
- + Log injury form for new entries

8.4 Workflow

- Player gets hurt → log via + Skrá meiðsli
- Status starts at 'injured', RTP stage 0
- Daily: physio assesses, advances stage when criteria met

- Verdict engine uses this in real-time — player's verdict on dashboard becomes 'Out — injury' / 'Rehab' / 'RTP' according to current status
- Marking cleared sets status='cleared', actual_return_date, and removes the injury banner from the dashboard

9. Underlying Signals (Reference)

All views above slice the same underlying signals. Quick reference for what each signal means and where it comes from.

9.1 STEN (Standardized Ten)

1–10 scale where 5–6 = personal average, 8+ = peak form, ≤ 2 = very low. Computed from z-score of today's check-in vs 28-day personal baseline. Used as the primary readiness number.

9.2 ACWR (Acute:Chronic Workload Ratio)

7-day cumulative load divided by 28-day weekly average. Sweet spot 0.8–1.3 (Gabbett 2016, Hulin 2014). > 1.5 = high injury risk zone. Computed separately for RPE-based load (internal) and Player Load (external).

9.3 Composite Load Concern (Outdoor)

0–1 score with bands none / low / moderate / high. Combines RPE ACWR (40%), GPS Neuromuscular Burden (35%), Metabolic Load Score (25%) when all three signals available. Falls back to 2-signal or 1-signal modes with confidence discount if data is sparse. Safety nets bump concern up: Residual MLI/Decel (3-day accumulation), Decel Burden, HID% fatigue trend, Accel:Decel ratio.

9.4 GPS Neuromuscular Burden (NBS)

0–1 sub-score that goes into the outdoor composite at 35% weight. Each Catapult GPS signal becomes a today/28d-baseline spike ratio, normalised against an alert threshold, then weighted:

Signal	Weight	Threshold (low → high)
HIR Spike (High Intensity Running)	34%	1.15 → 1.60
Decel Spike	26%	1.15 → 1.60
Density Stress Ratio	20%	1.10 → 1.35
Max Velocity Exposure	14%	1.02 → 1.12
Band 6 (sprint) Distance	6%	1.20 → 1.50

9.5 Indoor NBS (Catapult IMU-only)

Same purpose as GPS NBS but with FMP/IMU signals replacing GPS-derived ones (since GPS doesn't work indoors). Same weight distribution (34/26/20/14/6) with logical equivalents:

Indoor signal	Weight	Replaces (logical role)
FMP Dynamic High Spike	34%	HIR (high-intensity COD/accel/decel)

PlayerLoad Spike	26%	Density Stress (overall load)
IMA Total Spike	20%	Decel Spike (impact load)
FMP Dynamic Medium Spike	14%	Max Velocity (moderate intensity)
FMP Running High Spike	6%	Band 6 (linear running)

9.6 Indoor Composite Score (separate from Indoor NBS)

Different metric: a magnitude score (0–150+, 100 = personal baseline) shown on /coach/indoor-load. Uses 3 IMU-native components — Player Load 40%, Dynamic High % 33%, IMA total 27%. Indoor NBS is a concern probability; Indoor Composite is a magnitude index. Both real, both used, in different places.

9.7 Fatigue type

Fatigue type	Trigger	Meaning
mechanical_fatigue	GPS NBS ≥ 0.34 , Metabolic < 65	Body's mechanical systems under stress — sprints, decels
metabolic_fatigue	GPS NBS < 0.34 , Metabolic ≥ 65	Energy systems under stress — high-intensity running
global_fatigue	Both $\geq$ thresholds	Both mechanical and metabolic — most concerning
normal	Neither flagged	Within healthy range

10. Settings — Training Mode Toggle

On the Settings page (top nav → Admin → Settings), the Load Monitoring Mode card has a 3-state segmented control:

- 🤖 Auto (default) — per-player heuristic decides Indoor or Outdoor pipeline based on recent session counts. ≥ 3 indoor sessions/week = höll mode; otherwise outdoor.
- 🏟 Indoor — forces FMP / PlayerLoad / IMA pipeline. GPS metrics ignored in verdict explanation. Use during the hall season or for indoor sports.
- 🌿 Outdoor — forces GPS pipeline (HIR, velocity bands, sprint count). FMP indoor metrics ignored. Use during outdoor pitch season.

Iceland football pattern: most coaches set this twice per season (April flip to outdoor, October flip back to indoor). Basketball locks to Indoor automatically.

11. Coach Workflow Patterns

11.1 Daily morning check (5 min)

- Open dashboard → Today tab
- Read Daily Briefing one-sentence verdict + scan team status tiles
- If Top Concerns shows names, click each → read Why + Coach guidance + Question for player

- If team verdict is FULL with no concerns, you're done — run plan as normal

11.2 Pre-session 30-min review (10 min)

- Confirm Catapult sync from yesterday completed (header button shows sync state)
- Re-check any players flagged amber or red in the morning briefing
- If hall session: open Indoor Load Intelligence and look at the Team Heatmap
- If outdoor session: open Quadrant View at 7-day window, spot anyone in INJURY RISK quadrant
- Ask the players from Top Concerns the suggested check-in question before warmup

11.3 Weekly review (30 min, e.g. Mondays)

- Quadrant View at 14-day window — who shifted quadrants since last week?
- Decel Intelligence — anyone tip into Overload exposure or Underload risk?
- Indoor Load Intelligence Team Heatmap — sustained heavy load across 7+ days for any player?
- Open RTP tab → advance any rehab players who met stage criteria last week

11.4 Monthly retrospective (60 min)

- Injury Pattern Analysis — look at the team summary % preceded by warning
- Click each injury — were the warning signs there 7 / 14 / 21 days before?
- If "No prior signal" pattern: feedback into the monitoring approach (different STEN sensitivity, more frequent CMJ?)
- Adjust Composite Load thresholds with sport-science staff if needed

12. Sport-Science References

- Gabbett TJ. The training—injury prevention paradox: should athletes be training smarter and harder? British Journal of Sports Medicine 2016; 50(5):273–280.
- Gabbett TJ, et al. The athlete monitoring cycle: a practical guide to interpreting and applying training monitoring data. BJSM 2017; 51(20):1451–1452.
- Hulin BT, et al. Spikes in acute workload are associated with increased injury risk in elite cricket fast bowlers. BJSM 2014; 48(8):708–712.
- Vanrenterghem J, et al. Training load monitoring in team sports: a novel framework separating physiological and biomechanical load-adaptation pathways. Sports Medicine 2017; 47(11):2135–2142.
- McBurnie AJ, Harper DJ, Jones PA, Dos'Santos T. Deceleration training in team sports: another potential 'vaccine' for sports-related injury? Sports Medicine 2022; 52(1):1–12.
- Harper DJ, et al. High-intensity acceleration and deceleration demands in elite team sports. Sports Medicine 2019; 49(12):1923–1947.
- Halson SL. Monitoring training load to understand fatigue in athletes. Sports Medicine 2014; 44(Suppl 2):S139–S147.

13. Closing notes

Every view in this guide reads from the same underlying signals. They differ in which question they ask and which slice of the data they emphasise. A coach starting their morning on the

Today Command Center, drilling into individual players via the Decision Summary modal, and reviewing weekly via Quadrant + Decel Intelligence + Indoor Load is using all the relevant signals without ever needing to see raw Catapult exports.

MicroPulse is designed to be an explanation engine, not a black box. Every verdict and every flag has a Why line that traces back to the underlying signals. The coach is always the final decision-maker — the system surfaces what to consider, then trusts the coach's judgement.

MicroPulse — April 2026